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SUPERSEDING
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MILITARY SPECIFICATION

LOADER-TRANSPORTER, GUIDED MISSILE:
XM501E2 AND XM501E3
(OVERHAUL)

1. SCOPE

1.1 Scope. This specification covers overhaul of a full-tracked, self-propelled, tactical missile loader for loading and unloading Hawk Guided Missiles onto and from Hawk Missile System Launchers and Pallets.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS
Military

MIL-P-514	-	Plates; Identification, Instruction and Marking, Blank
MIL-H-5440	-	Hydraulic Systems, Aircraft Types I and II, Design, Installation, and Data Requirements for.
MIL-H-5606	-	Hydraulic Fluid, Petroleum Base, Aircraft Missile and Ordnance.
MIL-L-45779	-	Loader-Transporter, Guided Missile: XM501E2 and XM501E3, Processing for Storage and Shipment of.

FSC-1450

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STANDARDS
Military
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-130 - Identification Marking of U. S. Military Property.
MIL-STD-193 - Painting Procedures Tactical Vehicles (Tracked and Wheeled).
MIL-STD-461 - Electromagnetic Interference Characteristics Requirements for Equipment.
MIL-STD-642 - Identification Marking of Combat and Tactical Transport Vehicles.

DRAWINGS
Engineering List of Drawings (LD)
LD 8750307 - Loader-Transporter, Guided Missile, XM501E3
LD 9099200 - Loader-Transporter, Guided Missile, XM501E2

PUBLICATIONS
Engineering List of Parts List (LP)
LP 8750307 - Loader-Transporter, Guided Missile, XM501E3
LP 9099200 - Loader-Transporter, Guided Missile, XM501E2

Engineering List of Specifications (LS)
LS 8750307 - Loader-Transporter, Guided Missile, XM501E3
LS 9099200 - Loader-Transporter, Guided Missile, XM501E2

Master List of Supplementary Quality Assurance Provisions (QM)
QM 8750307 - Loader-Transporter, Guided Missile, XM501E3
QM 9099200 - Loader-Transporter, Guided Missile, XM501E2

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification and unless otherwise indicated, the issue in effect on date of invitation for bids shall apply:

MISSILE PURCHASE DESCRIPTION
MPD-7005 - Limit Environmental Criteria for the Anti-aircraft Guided Missile System - Hawk.

(Application for copies should be addressed to the U.S. Army Missile Command, AMSMI-IDS Redstone Arsenal, Alabama 35809).

3. REQUIREMENTS

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3.1 First overhauled vehicle. The first overhauled vehicle, which shall be fully representative of vehicles proposed to be furnished under the contract, shall be submitted to the Government for inspection to determine conformance to the requirements of this specification.

3.2 Workmanship and reliability. The workmanship shall be of quality to assure that the overhauled vehicle and components thereof are free from any defects that compromise, limit, or reduce the vehicle or component capabilities herein expressed. Vehicles overhauled in accordance with this specification shall be subjected to examinations and tests to prove vehicle capabilities to perform in accordance with the requirements specified herein.

3.3 Materials.

3.3.1 Qualified products list (QPL). With respect to all assemblies and component parts requiring product qualification in accordance with the pertinent documents cited in 3.4, and necessary in the construction of this vehicle, the contractor shall be responsible for using only those products which are listed (by part or drawing number as QPL items) on the QPL in the referenced specification or which have been approved for inclusion on such lists. Manufacturer's inspection records shall specifically list such component(s), name of supplier(s) (as listed on the QPL or in the approval letter), and number and date of the QPL from which the selection was made or date of approval letter when products are approved but not as yet listed on the QPL.

3.4 Overhaul. Vehicles shall be overhauled in accordance with procurement contract, this specification, and specified documents. When new parts are required, they shall conform to requirements listed in LD 8750307, LD 9099200, LP 8750307, LP 9099200, LS 8750307, LS 9099200, and referenced documents specified therein. All parts requiring identification shall be identified in accordance with MIL-STD-130.

3.4.1 Environmental. The loader shall perform satisfactorily when subjected to applicable environmental conditions specified in Purchase Description MPD-7005.

3.4.2 Fuel system. The fuel system shall maintain fuel supply to the engine when ascending and descending 60 percent grades in forward and reverse gear, and when vehicle is being operated on 30 percent side slopes, with each side of vehicle up slope.

3.4.2.1 Fuel tank and lines. Provisions shall be made to assure the internal cleanliness of fuel tank, lines, and connections prior to initial fueling of vehicle. Fuel tank, lines, and connections shall be free from leakage after installation and after break-in run.

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3.4.3 Hydraulic system. The loader hydraulic system shall conform to the applicable portions of MIL-H-5440.

3.4.3.1 Hydraulic reservoir and lines. Provisions shall be made to assure the internal cleanliness of hydraulic reservoir, lines, and fittings prior to initial filling of the hydraulic system.

3.4.3.2 Hydraulic fluid leakage. The loader hydraulic system specified herein shall evidence no sign of leakage through reservoir seals, bearing lubricant gaskets, joints, and manifold which operate at nominal hydraulic pressure under all conditions of operation, except at moving seals. The leakage at moving seals shall not exceed one drop per 25 cycles. Engine transmission and all suspension system seals shall prevent leaking of lubricants.

3.4.4 Exhaust system. There shall be no looseness or leaks in the exhaust system.

3.4.5 Adjustment mechanisms. All adjusting mechanisms shall be operational and maintain adjustment settings under all operating conditions.

3.4.6 Fasteners. All fasteners shall be positive in fastening.

3.4.7 Welding and burning. Welding and burning of any type or class shall be made only when, and to the extent, specifically authorized by the procuring activity.

3.4.8 Stowed equipment. When all on vehicle equipment (OVE) is stowed on the vehicle in the spaces provided material shall not interfere with the operation of vehicle and components in any manner. For stowing purposes, the manufacturer installed OVE as furnished by the vehicle contractor, and the depot installed OVE as furnished by the Government, shall be of the latest configuration and production.

3.4.9 Transportability. The overhauled loader, without missiles, shall be readily transportable by rail, motor transport, ship, air and by helicopter. It shall withstand, without functional damage, severe transport conditions, including cross-country travel over rough, mountainous, swampy roads, and by air transport in accordance with MPD-7005.

3.4.10 Structural interferences. All moving joints, fittings, hoses, cables, and structural members shall be free from detrimental interferences.

3.4.11 Missile transport. The loader shall transport 1, 2, or 3 missiles to whatever location is necessary within a Hawk missile system battery commensurate with the loader requirements as specified herein.

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3.4.12 Lifting and towing. The overhauled loader shall operate as a general lifting and towing device at a battery site, except as restricted by the operational limitations of the loader as specified herein.

3.4.13 Superstructure alignment. With the index boom properly indexed to the pallet or launcher, the loader hoisting beam latches shall be in proper alignment with the applicable missile latching points.

3.5 Break-in run (all vehicles).

3.5.1 Preparation. Prior to break-in run, the overhauled loader shall be completely assembled and all adjustments completed. The loader, including all components requiring lubrication, shall be lubricated and serviced as required using standard military products as shown on the lubrication drawings.

3.5.2 Engine warm-up. Prior to start of break-in run, or any time during the run when the engine has been stopped for a period of 15 minutes, the engine, when started, shall be warmed-up for at least 3 minutes and all engine indicator lights and gages shall indicate normal operating conditions.

3.5.3 Distance operated. Each overhauled loader, without missiles and with superstructure in normal transport position, shall be given a break-in run on smooth, level, hard-surfaced roads, except a section of the road shall be as shown in figure 1. The loader shall be operated as specified in table I, and over the section of road shown in figure 1 in forward and reverse gear.

Table I. Speeds and distance for break-in run

Division of run		Engine Speed RPM	Drive-selector position	Distance Miles
a	15	1500	H	0.5
b		2200	H	0.5
c		2000	R	50 feet
d		1500	D	0.5
e		2400	D	0.5
f		2000	R	50 feet

3.5.3.1 Reverse operation. Before operating in reverse (R), each time during the break-in run, the overhauled loader shall be brought to a stop, placed in neutral and the engine allowed to idle for not less than two minutes.

3.5.4 Condition after break-in run. After completion of the break-in run, there shall be no excessive grinding or backlash in gear trains; all suspension wheel bearings shall be correct or readjusted as required. There shall be no indication of damaged components or maladjustments that may cause faulty vehicle operation.

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3.6 Performance. Unless otherwise specified, the complete overhauled loader(s), unloaded, loaded, or with simulated load, serviced with specified fuel, lubricants, and fluids shall perform as specified herein. The vehicle(s), shall operate without evidence of excessive vibration; overheating of engine, powertrain, assemblies, and suspension system; abnormal noise, and ordors. There shall be no evidence of leakage of fuel, oil, water, lubricants, and hydraulic fluids. There shall be no malfunction during operation. Unless otherwise specified, loader(s) shall demonstrate performance requirements on relatively level, smooth, hard-surface roads.

3.6.1 Loader mobility. Unless otherwise specified, the loader with superstructure in transport position shall meet specified performance and operating requirements under all environmental conditions specified herein.

3.6.2 Power package. When power package is installed in accordance with applicable drawings, it shall function throughout all gear and speed ranges without loss of lubricants, or damage that may cause failure of loader.

3.6.3 Power train. The power train and associated controls shall operate throughout all speed and steering ranges without binding or leakages, loss of lubricants, grabbing, chattering and slippage when controls are applied to operate, stop, and hold vehicle.

3.6.4 Speed. With drive selector in drive (D) position, the loader shall start out with transmission in low range, then shift to transmission intermediate range, then shift to transmission high range, and shall maintain a speed of not less than 10 miles per hour (mph) for extended periods of time. With drive selector in hilly (H) position, loader shall operate at sustained speeds up to 5 mph without transmission shifting.

3.6.5 Creeping. With drive selector in D position and utilizing brakes as required, the loader shall be capable of smooth creeping at speed not to exceed 6 inches per second.

3.6.6 Acceleration. The loader shall accelerate from a standing start, with engine at idle, to 10 mph in not more than 5 seconds.

3.6.7 Climbing. The loaded vehicle shall climb, in both forward and reverse gear (drive selector in H and R respectively), a relatively smooth longitudinal 40-percent grade from a standing start on the grade. The loader, without missiles, shall be capable of the same performance on a relatively smooth 60-percent longitudinal grade.

3.6.8 Braking.

3.6.8.1 Stopping. The overhauled loader, traveling at 10 mph, shall stop within 10 feet from the point of brake application.

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3.6.8.2 Holding. With the vehicle loaded with intended load or a simulated load of equal size and weight, and on a 40 percent longitudinal slope, the service brakes shall hold vehicle stationary for not less than 3 minutes when headed up and down slope. With brakes locked for parking and all other holding devices inoperative, the loader shall be held stationary when headed up and down slope. On a relatively level, smooth, hard surface, the loader shall not move when engine is accelerated to a minimum of 1000 rpm, with drive selector in H position and both brakes fully applied.

3.6.9 Turning. The overhauled loader shall turn to the right and to the left, in forward or reverse, full 360-degree turns with the carrier remaining within a circle 40 feet in diameter.

3.6.9.1 Brake lever force. Brake lever force necessary to start the turn at all temperatures down to 32°F, shall be 40 pounds maximum. Maximum lever increase in lever force at all temperatures between 32°F and minus 65°F shall not exceed 20 percent. With all brakes released and steering levers in the full forward position (for "straight-ahead" or "straight-back" loader movement) and left untouched, loader shall maintain a relatively straight course when operated in the forward and reverse direction. When traveling at a speed between 5 and 10 mph, drift shall not exceed 3 feet in 100 feet.

3.6.10 Side slope. The overhauled loader shall transport at maximum speed a complement of three missiles on a smooth 30 percent side slope. With less than three missiles and load unbalanced, the loader shall operate at maximum speed in a downhill direction on a relatively smooth 18 percent side slope.

3.6.11 Trench crossing. The overhauled loader shall cross trenches up to 25 inches wide at reduced speeds as specified herein.

3.6.12 Vertical obstacle. The overhauled loader shall cross over vertical obstacles up to 10 inches high at reduced speeds as specified herein.

3.6.13 Blast hole. The overhauled loader shall negotiate a generally circular blast hole of 5 feet in diameter and 18 inches in depth at reduced speed as specified herein.

3.6.14 Rough terrain. The overhauled loader shall negotiate hard, boulder-strewn terrain, with maximum obstacle height of 5 inches, at speeds up to 5 mph. The loader shall negotiate undulating terrain (see figure 1) at speeds up to 10 mph as specified herein.

3.6.15 Operation in snow. The overhauled loader shall operate for extended periods of time in wet snow up to 12 inches in depth and perform as specified herein.

3.6.16 Fording. The overhauled loader, without use of special equipment shall ford a salt or fresh water crossing up to 20 inches in depth, including sinkage and wave height.

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3.6.17 Towing. The overhauled loader, without missiles, shall tow a load developing a minimum breakaway drawbar pull of 5,000 pounds and a continuous drawbar pull of 2500 pounds for a distance of not less than 1,000 feet at a minimum speed of 2 mph over relatively level and smooth, firm terrain.

3.6.18 Loader missile handling.

3.6.18.1 Hydraulic pressure at loader engine start-up. With starter switch of engine in starting position, engine turning over, and hydraulic load removed from engine, the hydraulic pressure registered on loader hydraulic oil pressure gage shall not exceed 50 psi. With engine running and starter switch released, (normal loader operating position), hydraulic oil pressure shall be as specified in 3.6.20.

3.6.18.2 Loader electrical system. The overhauled loader electrical system shall operate from a regulated 24 volt direct current (DC) supply and shall perform as specified herein under the range of environmental and service conditions to which the loader is exposed during operation. All electrical components shall be securely mounted, electrical connections shall be tight, and externally exposed electrical connectors shall have watertight seals. Cabling shall show no evidence of broken insulation, broken strands, fraying, and detrimental rubbing against other loader parts.

3.6.18.3 Master switch. With master switch "OFF" , no electrical power shall be provided to operate the loader electrical system. With master switch "ON", power shall be supplied to the elctrical system and the carrier and superstructure circuits shall operate as specified herein.

3.6.18.4 Ignition switch. With ignition switch at "ON" position, the engine starting and operating circuits for engine indicator lights, gages, missile latch and release indicator lights, and fuel pump shall be operable. With switch at "OFF" position, the operating circuits and fuel pump shall be inoperative.

3.6.18.5 Engine starting circuit. With overhauled loader in operating condition, properly serviced, and with drive selector in "N" position, starting circuit shall function as follows when starter switch is operated as follows:

(a) With the ignition switch "OFF" and the starter switch in the engine-starting position, the starter motor shall operate, but the engine shall not start.
(b) With the ignition switch at "ON" and the starter switch in the engine-starting position, the starter motor shall operate, and the engine shall start.

3.6.18.6 Lighting circuits. With light switch assembly on the instrument panel and adjusted as shown in table II, lights shall operate as specified therein.

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Table II. Instrument Panel Lights

Switch setting and operating conditions	Light condition
1. Main light switch at "OFF"	No carrier lights operate
2. Main light switch at "SERVICE DRIVE"	Head and tail lights on.
a. Dimmer switch operated	Headlight beam changes.
b. Brakes applied	Stoplight on.
c. Brakes released	Stop light off.
3. Main light switch at "STOP LIGHT"	Head lights off, tail lights off.
a. Brakes applied	Stop light on.
b. Brakes released	Stop light off.
4. Main light switch at BLACKOUT (BO) marker	BO marker and BO tail lights on.
a. Brakes applied	BO stop light on.
b. Brakes released	BO stop light off.
5. Main light switch at (BO) drive	BO drive light on BO marker and BO tail lights on.
a. Brakes applied	BO stop light on.
b. Brakes released	BO stop light off.
6. Main light switch at any setting except "OFF" and panel light switch at settings:	Lights operate as specified in 2, 3, 4, or 5 as applicable.
a. "OFF" position	Instrument panel lights off.
b. "PARK"	Instrument panel lights on.
c. "DIM"	Instrument panel lights dim.
d. "PANEL BRIGHT (BRT)"	Instrument panel lights bright.

3.6.18.7 Safety switch. With safety switch "ON", and light switch assembly in normal position, main light switch (see table II) shall be operable only to the "OFF" or (BO) marker position. Main light switch shall be turnable to "SERVICE DRIVE", "BO DRIVE", or "STOP LIGHT" positions as specified in table II only when safety switch is simultaneously held in unlock position.

3.6.18.8 Engine indicator circuits. Engine indicator gages and lights shall indicate as shown in table III.

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Table III. Indicator gages and lights

Indicator gage or light	Indication
a. With ignition switch "OFF"	
1. Battery generator (BATT-GEN) indicator	Actual battery voltage.
2. Other gages and indicator lights	Lights off gages do not register.
b. With ignition switch "ON" engine not running	
1. (Batt-gen) indicator	Actual battery voltage.
2. Fuel level gage	Correct quantity of fuel.
3. Engine and transmission (eng and trans) low oil press warning lights	On.
4. Transmission, differential, hydraulic, (Trans, diff, hyd) and water high temperature (temp) warning lights	Off (unless oil, hydraulic or water in high temperature range, or unless indicator lights tests switch is depressed).
5. Indicator lights test switch actuated	All indicator warning lights on.
c. Engine running at fast idle	
1. (BATT-GEN) indicator	Actual generator (voltage regulator) voltage.
2. Fuel level gage	Correct quantity of fuel.
3. All indicator warning lights	Off, except as specified in (b) 4 above
4. Indicator lights test switch actuated	All indicator warning lights on.

3.6.18.9 Auxiliary power outlet. Auxiliary 24 volt DC power fused at 15 amperes (amps) shall be provided at the utility outlet.

3.6.18.10 Lights. Normal headlights and taillights, blackout headlights and taillights and all instrument panel lights shall conform to applicable drawings. All electrical contacts and connections shall maintain positive contact under all loader operating conditions.

3.6.19 Controls.

3.6.19.1 Engine, transmission and electrical equipment controls. The engine, transmission, and electrical controls shall be capable of positive operation at all times and under all operating conditions.

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3.6.19.2 Suspension system lockout controls. With the suspension lockout control in lockout position, the right and left front and center wheels of the suspension system shall be in lockout and the overhauled loader shall load and unload a pallet or operate as a crane. With the controls returned to the unlocked position the suspension system shall return to normal operation. Suspension lockout control shall be engaged or disengaged with superstructure in transport position.

3.6.19.3 Brake controls. With brake control louvers fully released or locked, top of brake lever handles shall be in alinement with each other within one-half inch. The force required to lock the brakes shall not vary between the two levers by more than five pounds. With each brake lever in fully locked position, there shall be a minimum of two notches of travel remaining.

3.6.19.4 Superstructure controls. With the superstructure manual operating controls in "NEUTRAL" position, the superstructure shall remain approximately as positioned. Operation of the controls shall provide smooth positive actuation of the superstructure throughout all ranges of azimuth, elevation, roll, and extension under all operational conditions.

3.6.19.5 Engine starting lockout control. With the drive selector lever in any position other than neutral (N) the engine starter electrical circuit shall be inoperative.

3.6.20 Hydraulic system pressure.

3.6.20.1 Hydraulic pump. The operating pressure of the engine driven hydraulic pump shall be 3,000 pounds plus 200 minus 100 pounds per square inch gage (psig) when using oil conforming to MIL-H-5606. The maximum system pressure shall be limited to 3,400 psig by a system relief valve.

3.6.20.2 Proof pressures. The hydraulic system shall withstand proof pressures as specified in MIL-H-5440 without damage.

3.6.21 Superstructure operation (normal loader operation). The loader superstructure, empty or loaded with 1, 2, or 3 missiles, shall operate smoothly and satisfactorily under operating conditions as specified herein. With superstructure adjusted for normal loader operation as specified in table IV and with engine running, superstructure shall respond as specified herein when superstructure control levers are moved from their respective neutral positions as specified herein.

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Table IV. Normal loader operation

Lever and lever motion	Superstructure movement	
	Motion	Range of travel
a. ELEVATION - ROLL		
1. Forward and back	Index boom respectively depresses and elevates	700 mils (Minimum) of elevation and depression.
2. Right or left (Control centers when released)	Index boom rolls respectively right or left	267 mils (Minimum) in either direction from center level position, except as limited by 3.6.21.4
b. AZIMUTH		
1. Clockwise or counter clockwise (Control centers when released)	Main support rotates in azimuth respectively clockwise or counter-clockwise	267 mils (Minimum) in either direction from center position, except as limited by 3.6.21.4.
c. EXTENSION		
1. Forward or back	Combined link assembly and transfer arms respectively extend and retract	Not applicable

3.6.21.1 Missile release circuits. The hoisting-beam latch-release solenoid shall not operate when the missile-latch-release switch located on instrument box cover is in the normal (open) position.

3.6.21.2 Superstructure operating speeds. With overhauled loader on a relatively level, smooth, firm surface, with control levers at maximum settings and engine speed as specified in table V, the superstructure, loaded with three missiles, shall operate as specified in the table.

3.6.21.3 Superstructure settling. With three missiles on the overhauled loader, the elevation and extension cylinders extended to any position within their respective ranges, the elevation and extension valves in neutral, and with engine off, superstructure shall not settle more than one inch (measured vertically at missiles) in one minute. (Caution: If transfer arms are raised more than 15 degrees, loader must be engaged with pallet or launcher to prevent overturn.)

3.6.21.4 Azimuth and roll-control limitations. Azimuth and roll controls shall be operative when the missile is more than 10 inches above the transport

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position and shall be inoperative when the missile is less than 2 inches above the transport position.

Table V. Superstructure operating speeds

Operation	Engine RPM	Traverse Time (entire range) seconds
a. ELEVATION CYLINDER		
1. Elevate	2,000 - 2,500	10.0 to 21.0
2. Depress	2,000 - 2,500	10.0 to 21.0
b. ROLL CYLINDER		
1. Full right to full left	2,000 - 2,500	15.0 to 21.0
2. Full left to full right	2,000 - 2,500	15.0 to 21.0
c. AZIMUTH CYLINDER		
1. Rotate from full right to full left	1,500 - 2,000	4.0 to 12.0
2. Rotate from full left to full right	1,500 - 2,500	4.0 to 12.0
d. EXTENSION CYLINDER		
1. Extend (loader indexed to pallet on ground)	3,000 - 3,200	*15.0 to 30.0
2. Retract (loader indexed to pallet on ground)	3,000 - 3,200	*15.0 to 30.0

*Entire range includes reduced pressure travel of missiles on pallet.

3.6.21.5 Reduced pressure control. Extension and retraction of the superstructure during the final phase of extension and the initial phase of retraction, shall be accomplished under reduced hydraulic pressure not to exceed 400 PSI by energizing the solenoid valve in the hydraulic system extension control loop. The solenoid valve shall energize under conditions (a) and (b) below, and de-energize under condition (c) and remain energized under condition (d).

(a) Index and transfer arm limit switches are actuated (plungers extended).

(b) During extension.

1. Without missiles, the solenoid valve shall energize at least 0.5 inch before the loader hoisting beam forward snap-in latch seats in the upper-aft missile

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pocket, and not before the transfer arm separates from the intermediate link on the limit switch side.

2. With missiles, the solenoid valve shall energize at least 0.5 inch before the applicable lower-aft missile pocket seats against the aft pallet hook, and not before the transfer arm separates from the intermediate link on the limit switch-side.

(c) During retraction.

1. Without missiles, the solenoid valve shall de-energize after the loader hoisting beam forward snap-in latch has moved at least 0.5 inch from the upper aft missile pocket and prior to closure of the gap between the transfer arm and the intermediate link on the limit switch side.
2. With missiles, the solenoid valve shall de-energize after the lower aft missile pocket has moved at least 0.5 inch from the aft pallet hook and prior to closure of the gap between the transfer arm and the intermediate link on the limit switch side.

(d) During extension and retraction.

1. With any combination of 1, 2, or 3 missiles and with the missile(s) latched to the pallet and to the loader simultaneously, the solenoid valve shall remain energized and there shall be no permanent deformation or damage to the loader when extension control lever is operated in the extended direction.

3.6.21.6 Manual latching of missiles. Positive latching and locking of 1, 2, and 3 missiles to superstructure hoisting beam(s) shall be accomplished by manually operating applicable latching handle(s) to the latched position. In the latched position, missile latches shall fully extend and remain locked until rear hoisting beam latches are retracted by manual or electrical release.

3.6.21.7 Manual release or missiles. Latched missiles shall release from superstructure hoisting beams when the applicable latch release solenoid plunger is manually depressed (retracting aft latch), and when missiles are at rest on pallet saddles so the weight of the missiles shall be completely off hoisting beams. In the released condition, aft latches shall be fully retracted.

3.6.21.8 Forward missile supports. Outboard forward missile supports, shall rise from the stowed position (at rest on loader fenders) to the normal loader position (for support of missiles during loader transport) by rotating support lever handles from the full forward positions to full back and vice versa.

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3.6.21.9 Rear center missile support. The rear center missile support and safety bar shall fold down to the stowing position for loader shipment and for crane operation.

3.6.22 Superstructure electrical system.

3.6.22.1 Main electrical cable. With main electrical cable assembly connected between superstructure and distribution box for normal loader operation, superstructure electrical circuits shall operate as specified herein. With superstructure cable disconnected and attached to dummy connector, superstructure electrical circuits shall not function.

3.6.22.2 Superstructure circuits (normal loader operation). With superstructure adjusted for normal loader operation with missiles, and with master switch "ON", superstructure circuits shall function as specified herein.

3.6.22.3 Missile latch and release circuits. With ignition switch "OFF", the "missile-latched" and "missile-released" indicator lights shall not illuminate. With ignition switch "ON" and with loader engine and superstructure operating or at rest, "missile-released" and "missile-latched" lights, the indicator-lights-test switch and dim bright control shall function as specified herein.

3.6.22.3.1 Missile latch and release lights. Except as specified, the right, center, or left "missile-released" light (red) shall illuminate when missile latches in the corresponding (right, center, or left) hoisting-beam assembly are in the released condition. Right, center, or left "missile-latched" light (green) shall illuminate when missile latches in the corresponding hoisting beam assembly are in latched condition.

3.6.22.4 Indicator-light-test switch. All "missile released" and "missile latched" lights, whether individually "ON" of "OFF", shall illuminate when the "indicator-lights-test switch" is actuated (switch button held in full depressed position). Indicator lights shall return to their initial condition when switch button is released (switch returned to normal "OFF" operating position).

3.6.22.5 Dim bright control. Rotation of dim "brt" control from "dim" (maximum counterclockwise position) to "brt" (maximum clockwise position) shall brighten all illuminated "missile-released" and "missile latched" lights and vice versa. With control at "dim," missile-released lights shall be readily visible in the illuminated condition (red), during complete darkness. With control at "brt", illumination of "missile-released" lights shall be readily visible in bright sunlight.

3.6.22.6 Electrical release of missiles. With 1, 2 or 3 missiles latched to loader, operation of missile "latch release" to momentary "ON" (up) position shall electrically release the one or more selected missiles from the loader when the following conditions are met:

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(a) Conditions for energizing latch release solenoid(s) applicable to missile(s) selected for release shall be as specified herein.
(b) Weight of missile(s) shall be off loader hoisting beam assembly housing(s) (full weight of missiles carried by pallet saddle(s) or other surface receiving released missile(s).

3.6.22.6.1 Latch release solenoids. With loader hoisting beam assemblies in the "missile-latched" condition as specified herein, energizing "latch release" solenoids shall retract latch locks so that spring-loaded latching assemblies will return to the "missile released" condition when weight of missiles is not suspended from latches. Solenoids shall energize only when the following conditions are met:

Master switch "ON"

(a) Index and transfer arm limit switches are actuated (plunger extended).
(b) Applicable circuit breakers are on (toggle "up" position).
(c) Missile "latch release" switch is actuated to "ON".

3.6.22.7 Index boom and transfer arm limit switches. Latch and release solenoids shall not energize unless series connected index boom-intermediate link limit switch and transfer arm-intermediate link limit switch are actuated (plungers extended). Index boom-intermediate link limit switch shall actuate only when superstructure is extended until link assembly is raised from index boom trunnion beyond a point where the limit-switch plunger is extended. Transfer arm-intermediate link limit switch shall actuate only when transfer arm is separated from the intermediate link assembly beyond a point where the limit-switch plunger is extended.

3.6.23 Loading and unloading missiles. The overhauled loader shall load and unload missiles under the following conditions:

(a) With the launcher positioned on slopes up to 10 degrees and with the loader on the same slope, the loader shall be capable of indexing to the launcher and loading or unloading the launcher with 1, 2, or 3 missiles. When loading or unloading less than 3 missiles, the missiles may be guided by hand if necessary.

(b) With the pallet mounted on an XM-390 type trailer, positioned on slopes up to 10 degrees and with the loader on the same slope, the loader shall be capable of indexing to the pallet and loading and unloading the pallet with 1, 2, or 3 missiles. When loading or unloading less than 3 missiles, the missiles may be guided by hand if necessary.

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NOTE: The trailer mounted pallet must be approximately leveled when the trailer pintle is down slope or when located on a side slope. When the trailer pintle is up slope the trailer bed must be approximately parallel to the ground.

(c) With the pallet on the ground and positioned on slopes up to 10 degrees and with the loader on the same slope, the loader shall be capable of indexing to the pallet and loading and unloading the pallet with 1, 2, or 3 missiles. When loading or unloading less than 3 missiles, the missiles may be guided by hand if necessary.

3.6.24 Loading and unloading pallet. With the suspension locked, the loader, either empty or with missiles aboard, shall be capable of engaging an empty pallet and transferring the pallet from an XM-390 type trailer to the ground, and from the ground to an empty XM-390 type trailer.

3.6.25 Crane.

3.6.25.1 Crane operation. With the suspension locked, the overhauled loader, when fitted with crane accessories and adjusted for crane operation, shall function as a crane on slopes up to 10 degrees commensurate with the loader crane operating requirements as specified herein. All moving joints, fittings, hoses, electrical cables, stays, and structural members shall be free from binding or detrimental interference during operation.

3.6.25.2 Crane controls. With loader engine operating, speed and direction of cable travel and swing and elevation of crane boom shall be controlled by operation of the superstructure hydraulic controls specified herein.

3.6.25.3 Roll lockout. With roll lockout lever placed in "crane" position, roll control shall be inoperative.

3.6.25.4 Hoisting cable. Operation of extension control lever forward and backward shall respectively raise and lower the maximum loads as specified herein.

3.6.25.5 Boom swing. With maximum loads as specified herein applied to boom, operation of "azimuth" control lever clockwise and counterclockwise shall respectively swing crane boom (ramp beam) clockwise and counterclockwise.

3.6.25.6 Boom elevation. With maximum loads as specified herein applied to boom, operation of elevation control lever forward and back shall respectively elevate and depress the crane boom.

3.6.25.7 Hoisting hook position. With the crane rigged without cable stay and load, the hoisting hook shall be positioned (relative to the ground) as specified below:

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(a) With the crane boom (ramp beam) fully depressed (elevation cylinder fully extended) and the transfer arm sockets seated on the index boom trunnion (hoisting cable fully extended), the lowest point of the hoisting hook shall not be higher than 12 inches above the ground.
(b) With the crane boom fully elevated (index boom resting on the rear missile support) and the extension cylinder fully extended (hoisting cable fully retracted) the lowest point of the hoisting hook shall be not lower than 150 inches above the ground.

3.6.25.8 Rated capacity. The loader, in a non-mobile condition on slopes up to 10 degrees, shall maintain stability when lifting a minimum load of 2500 pounds. The loader, in a mobile condition on slopes up to 10 degrees and operating at reduced speed over unimproved terrain, shall maintain stability when lifting a load of 1600 pounds. When operated at either condition, there shall be no permanent deformation or damage to the loader.

3.6.26 Radio interference suppression. Unless otherwise specified (see 6.2), the complete loader shall be suppressed to prevent interference to high-sensitivity radio communication equipment in the frequency range of 0.15 megacycles (mc) to 500 mc, in accordance with the requirements of Class III of MIL-STD-461.

3.7 Weight. Weight of the completely assembled loader, serviced with hydraulic fluid conforming to MIL-H-5606 but excluding missiles, fuel, storage items, crane installation and an operator, shall not exceed 5,500 pounds.

3.8 Painting, marking and data plates.

3.8.1 Painting. Unless otherwise specified, the exterior and interior of loader, components, assemblies, subassemblies, and parts that require painting shall be prepared and painted in accordance with MIL-STD-193 and applicable vehicle drawings.

3.8.2 Marking. Marking shall be in accordance with MIL-STD-642.

3.8.3 Name, shipping, and identification data plates. The information to be included shall be in accordance with applicable drawings or as specified by the procuring activity. The plates shall meet the performance requirements of MIL-P-514.

3.9 Fire extinguisher. When mounted in accordance with the drawings, the extinguisher shall be readily accessible and shall operate without malfunction.

3.10 Workmanship. The workmanship employed by the overhaul contractor of the loader shall be of such quality and consistency as to minimize degradation of the expected performance and design reliability. The overhauled loader shall be free from such defects as breaks, bends, cracks, deep cuts, rust, chipped or

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spotty paint, damaged fastenings, looseness of attaching parts, misalinement, binding and detrimental interference of parts. All bolted joints shall have reasonably flat mating surfaces that are free from burrs, weld spatters, and other significant surface irregularities.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Contractor's quality control system. Unless otherwise specified by the procuring activity, the contractor shall provide and maintain an effective inspection and quality control system acceptable to the Government covering the supplies under the contract. A current written description of the system shall be submitted to the contracting officer prior to initiation of production. The contractor will not be restricted to the inspection station or to the method of inspection listed provided that an equivalent control is included in the approved quality control procedure. The contractor shall notify the Government of and obtain approval for any change to the written procedure that might affect the degree of control required by this specification or other applicable documents referenced therein.

4.1.2 Parts and assemblies. Parts and assemblies shall be inspected in accordance with requirements of QM 8750307, QM 9099200, LD 8750307, LD 9099200, LP 8750307, LP 9099200, LS 8750307, LS 9099200, and referenced documents specified therein.

4.1.3 Government verification. All quality assurance operations performed by the contractor will be subject to Government verification at unscheduled intervals. Verification will consist of (a) surveillance of the operations to determine that practices, methods, and procedures of the inspection plan are being properly applied, and (b) Government product inspection to measure quality of product offered for acceptance. Deviation from the prescribed or agreed-upon procedures, or instances of poor practices which might have an effect upon the quality of the product, will immediately be called to the attention of the contractor. Failure of the contractor to promptly correct deficiencies shall be cause for suspension of acceptance until corrective action has been made or until conformance of product to specifications herein has been demonstrated.

4.1.4 Qualified products. The contractor's inspection records shall be checked to determine contractor conformance to 3.3.1.

4.2 First overhauled vehicle inspection.

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4.2.1 Special process inspection. A special process inspection shall be conducted by the Government during overhaul of the first vehicle (see 3.1) to evaluate conformance of materials and workmanship as specified herein. Repair procedures used to correct damages and wear shall be evaluated.

4.2.1.1 Vehicle frame inspection. All weldments shall be inspected prior to covering the weldments (see 3.4.7) with paint or other pertinent components or assemblies.

4.2.2 First overhauled vehicle inspection. Subsequent to the contractor's inspection of the first completed vehicle, the Government shall inspect this vehicle at the contractor's plant to determine conformance to contract and specification requirements.

4.2.2.1 Corrections. Corrections found necessary as a result of the above inspection shall be made by the contractor.

4.2.3 Initial overhauled vehicle tests. After completion of the first over-hauled vehicle inspection, another vehicle from the first month's overhaul or one of the first ten shall be subjected to examination (see 4.4.1) and tests as specified in 4.5 (a) and (b). Subsequent to examination and tests, the selected vehicle shall be subjected to the 250 hour test specified in Table VI and all applicable tests specified in Table VIII. These tests shall be performed by the Government, at a site approved by the Government, which will require a period of 90 days. Delays caused by vehicle break-down due to poor quality of workmanship or material, or failure of the contractor to furnish repair parts as required to adequately support the test or to comply with specifications or drawing requirements shall not be the basis for adjustment of the contract performance dates or delivery schedule or contract price.

Table VI. 250 hour test

A. Cycle breakdown for loader operation	
1. Missile transfer	70 hours
2. Missile transport	70 hours
3. Miscellaneous	60 hours
B. Cycle breakdown for mobile crane	
1. Load handling	12 hours
2. Maneuvering	12 hours
3. Miscellaneous	12 hours
C. Cycle breakdown for use as utility vehicle	
1. Maneuvering	6 hours
2. Miscellaneous	5 hours

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4.2.3.1 Test failure. Failure of the vehicle to comply with any of the requirements specified or any deficiency of a workmanship or materials nature found on the vehicle during or as a result of the 250 hour test, shall be cause for rejection of the vehicle. Further, the Government will refuse to continue acceptance of production vehicles until the contractor has taken corrective action to eliminate the deficiency. In addition, correction of the deficiency must be accomplished by the contractor at no cost to the Government. Any deficiency found as a result of the 250 hour test shall be considered prima facie evidence that all vehicles already accepted prior to completion of the 250 hour test are similarly deficient unless evidence satisfactory to the contracting officer is furnished by the contractor that they are not similarly deficient. Such deficiencies shall be corrected by the contractor at not cost to the Government.

4.3 Break-in. To determine conformance to 3.5.1, 3.5.2, and 3.5.3 each loader shall be lubricated and serviced prior to the break-in run. The run shall be on relatively level, smooth, hard-surface roads, and for the distances specified in table I. The loader shall be operated in reverse after each division as specified in 3.5.3.1. The undulating portion of the test shall be accomplished twice in each direction during division "a" and during division "b" as specified in Table I.

4.3.1 Condition after run. After the break-in run the loader shall be run at idle speed for a period of not less than 5 minutes, and inspected to determine conformance to 3.5.4.

4.4 Inspection provisions.

4.4.1 Examinations. Visual, dimensional, and primary functional examination shall consist of examination of the vehicle for conformance to the applicable drawings and this specification. Examination shall be performed against the classification of defects and with the acceptable quality levels shown in Table VII. The following constitutes a part of the classification of defects (see 6.5):

(a) Any drip constitutes a major defect when the vehicle has been standing idle and the components are at ambient temperature.
(b) Any droplet that occurs at a static fit, metal to metal or gasket combination shall constitute a major defect.

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Table VII. Classification of defects

Characteristics		Major AQL 25 Defects/100 units	Method of Inspection
Engine	101	Malfunction, leaks, (see 6.5 c & d) mounting improper, damage	Visual and Functional
Power train	102	Malfunction, leaks (see 6.5 c & d) mounting improper, damage.	Visual and Functional
Fuel system	103	Malfunction, damage, leaks, (see 6.5a).	Visual and Functional
Fuel, and oil lines, valve and hose	104	Defective, clearance improper, malfunction leaks (see 6.5a), fuel cut off valve damaged.	Visual and Functional
Exhaust system components	105	Damage, leaks.	Visual and Functional
Cooling system components	106	Malfunction, damage, leaks (see 6.5d), unusual noise.	Visual and Functional
Electrical system components	107	Improper installation, loose connections and components, safety wiring missing. Wire has excessive bends, pinches, wiring damage, defective cabling, frayed chafed, broken, burned, insulation damaged.	Visual and Functional
Suspension system components	108	Damage, improper clearance or alinement, welding defects, leaks, improper assembly, installation, bolt or nut torque and track tension.	Visual and Functional
Mating flanges	109	Surfaces not flat for mating when bolted or screwed together.	Visual
Hydraulic system	110	Hydraulic lines dented, twisted, buckled, excessive bend, chafed or excessively worn. Leakage (see 6.5d), hydraulic oil contaminated.	Visual and Functional

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Table VII. Classification of defects (Con't)

Characteristics		Major AQL 25 defects/100 units	Methor of Inspection
Brake lever	111	Top of brake levers out of alinement when released or locked. Brake lever travel incorrect when fully locked.	Visual and Functional
Quick release pins	112	Quick release pins do not fit holes, bent or damaged.	Visual and Functional
Center rear missile support	113	Will not adjust to stowed position and vice versa.	Visual and Functional
Outboard forward missile	114	Will not adjust from stowed positions to the missile transport position and vice versa.	Visual and Functional
Angle of forward missile	115	Angle of missile transport supports incorrect.	Visual
Inboard edges of index tips	116	Not parallel.	Visual
Centerline of hoisting beam housings	117	Not parallel and centerline of each housing not coincident with centerline of its latches, distance between adjacent centerlines of housings.	Visual
Torque on latching handles	118	Effort to latch.	Visual and Functional
Transfer arm limit switch plunger	119	Plunger does not extend beyond striker face.	Visual
Index boom limit switch	120	Ineffective travel of plunger.	Visual and Functional
Holes in extension cylinder support bracket	121	Holes improper size and location.	Visual
Holes in index forks	122	Holes improper size and location.	Visual

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Table VII. Classification of defects (Con't)

Characteristics		Major AQL 25 defects/100 units	Method of Inspection
Parts missings, improperly assembled and installed, hose, damaged or defective	123	Lockwashers, locknuts, locking wires, cotter pins, screws, bolts nuts, washers, rivets, captivating chains, wiring clips grommets, ground straps, missing grease fittings, cracks, breaks, tears, deep cuts, dents, bends, or structural deformation.	Visual
Paint	124	Application and color improper	Visual
Nameplates	125	Missing, not properly fastened to loader, incorrect or illegible.	Visual
Identification, operational, warning, and caution markings	126	Missing, incorrect, incomplete, illegible or not correctly located.	Visual
Loader	127	Weight of loader in excess of 5500 pounds.	Weighing Device
Instrumentation, switches, warning indicating, and safety devices	128	Malfunction, damage.	Visual and Functional
Missile circuit release test	129	Malfunction.	Functional
Superstructure circuit test	130	Malfunction.	Functional
Suspension system lockout test	131	Malfunction.	Functional
Missile latches	132	Improper latching	Visual and alignment fixture
Superstructure alinement	133	Misalinement of any element of the superstructure.	Alinement Fixture.

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Table VII. Classification of defects (Con't)

Characteristics	Minor AQL 150 Defects/100 units		Method of Inspection
Parts	201	Improperly assembled, loose, damaged or defective, also screws, bolts, nuts, washers, rivets, wiring clips, and grommets missing, incorrect, damaged defective, or improperly installed (not affecting loader functional performance).	Visual
Defects not affecting functional performance	202	Cracks, breaks, tears deep cuts, dents, bends, bulges, or structural deformation (not affecting loader functional performance).	Visual
Loader not clean	203	Dirt, mud, oil, grease, metallic chips, or other loose, or attached foreign particles or substances present.	Visual
Welding	204	Welding warped or of poor appearance	Visual
Lubrication	205	Lubrication fittings not as required	Visual
Identification, operational, warning and caution product markings	206	Damaged or not applied as specified (still useable).	Visual
Hoisting beam striping	207	Not striped as specified or striping damaged.	Visual
Azimuth index marks and roll index mark	208	Marks missing, illegible or incorrectly located.	Visual
Loader drain	209	Damaged or defective.	Visual
Pintle assembly	210	Damaged, defective, improperly installed, improper operation.	Visual
Stowage box	211	Stowage box not as specified, damaged, latches inoperable or wrong location	Visual

4.4.2.1 Unclassified defects. All defects having no effect on function, safety, interchangeability, and life of vehicle, but are considered departures from good workmanship will be noted in writing.

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4.4.2.2 Recurring major deficiencies. A major deficiency (see 6.6) is recurring when the same defect occurs more than once in the same sample or when the defect occurs in two successive samples. A major defect may be considered recurring when the historical inspection records ("P" chart or approved equivalent) reflect such a condition. Recurring major deficiencies shall be cause for the entire lot or lots to be inspected for the recurring deficiencies. The deficiencies shall be corrected when found.

4.4.2.3 Recurring minor deficiencies. A minor deficiency (see 6.6) is recurring if it occurs more than twice in the same sample or when the defect occurs in four successive samples. Recurring minor deficiencies shall be cause for the entire lot or lots to be inspected for the recurring deficiencies. Correction of all deficiencies shall be accomplished prior to acceptance.

4.5 Classification of tests.

(a) Acceptance test 4.5.1 through 4.5.1.30.
(b) Control test 4.5.2 through 4.5.2.5.
(c) Comparison tests 4.5.3 through 4.5.3.16.

4.5.1 Tests.

4.5.1.1 Acceptance tests. To determine conformance to 3.6, after pre-test run (see 3.5) each loader selected shall be operated for a distance of 1 mile or more and shall be inspected as specified in 4.4.1. Acceptance tests shall include all acceptance tests specified at the place of manufacture in table VIII. Loaders shall be completely assembled and serviced, with fuel, lubricants, and fluid as specified herein. When specified (see 6.2), loader shall carry simulated missile payload. Operation shall be on relatively level, smooth, hard-surfaced road, except the road shall include a section as shown in figure 1, to test the functioning of suspension system and shock absorbers. Prior to actual tests, engine shall be operated between 650 and 750 rpm to warm engine and lubricants to the operating temperature. The loader shall be stopped with the engine idling for not less than 30 seconds and then operated in reverse for not less than 50 feet. To facilitate examination, observation, and function of certain areas, this test may be conducted without the floor plates, personnel seat cushions, OVE and affixed decals. After tests, the loader shall be examined for evidence of fuel or lubricants leakage, or other deficiencies.

4.5.1.1.1 Test failure. If the vehicle fails to pass any acceptance test specified herein, the Government inspector shall stop acceptance of subsequent lots until evidence has been provided by the contractor that corrective action has been taken.

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Table VIII. Location for tests

	Paragraph Reference Number	Place of Manufacture	Government Proving Ground
Acceptance tests	4.5.1.1	X	X
Fuel and hydraulic system check	4.5.1.2	X	X
Fuel system leakage check	4.5.1.3	X	X
Hydraulic fluid leakage check	4.5.1.4	X	X
Exhaust system check	4.5.1.5	X	X
Adjustment mechanisms check	4.5.1.6	X	X
Fasteners, and light check	4.5.1.7	X	X
Welding and burning check	4.5.1.8	X	X
Structural interferences check	4.5.1.9	X	X
Link assembly and transfer alinement check	4.5.1.10	X	X
Loader mobility tests	4.5.1.11	X	X
Power package and power train test	4.5.1.11.1	X	X
Speed tests	4.5.1.11.2	X	X
Creeping test	4.5.1.11.3	X	X
Acceleration test	4.5.1.11.4	X	X
Climbing test	4.5.1.11.5	X	X
Stopping test	4.5.1.11.6	X	X
Holding test	4.5.1.11.7	X	X
Turning test	4.5.1.11.8	X	X
Hydraulic pressure at loader engine start-up test	4.5.1.12	X	X
Loader electrical system check	4.5.1.13	X	X
Controls check	4.5.1.14	X	X
Hydraulic system test	4.5.1.15	X	X
Superstructure operation (normal loader operation) test	4.5.1.16	X	X
Missile release circuits test	4.5.1.16.1	X	X
Superstructure operating speeds test	4.5.1.17	X	X
Superstructure settling test	4.5.1.18	X	X
Azimuth and roll controls limitation test	4.5.1.19	X	X
Reduced pressure control test	4.5.1.20	X	X
Manual latching of missiles test	4.5.1.21	X	X
Manual release of missiles test	4.5.1.22	X	X
Foward missile support test	4.5.1.23	X	X
Rear center missile support test	4.5.1.24	X	X
Superstructure electrical system check	4.5.1.25	X	X
Electrical release of missiles test	4.5.1.26	X	X
Latch release solenoid test	4.5.1.26.1	X	X
Index boom and transfer arm limit switches test	4.5.1.27	X	X

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Table VIII. Location for tests (Cont'd)

	Paragraph Reference Number	Place of Manufacture	Government Proving Ground
Loading and unloading missiles test	4.5.1.28	X	X
Loading and unloading pallet test	4.5.1.29	X	X
Painting, marking, and data plates check	4.5.1.30	X	X
Control test	4.5.2	X	X
Stowed material check	4.5.2.3	X	X
Missile transport test	4.5.2.4	X	X
Side slope test	4.5.2.5	X	X
Comparison test	4.5.3	X	X
Environmental tests	4.5.3.2		X
Fuel system functional test	4.5.3.3		X
Transportability test	4.5.3.4		X
Lifting and towing tests	4.5.3.5		X
Brake lever force low temperature test	4.5.3.6		X
Trench crossing test	4.5.3.7		X
Vertical obstacle test	4.5.3.8		X
Blast hole test	4.5.3.9		X
Rough terrain test	4.5.3.10		X
Operation in snow test	4.5.3.11		X
Fording test	4.5.3.12		X
Towing test	4.5.3.13		X
Loading and unloading missiles test	4.5.3.14		X
Crane operation test	4.5.3.15		X
Crane control test	4.5.3.15.1		X
Roll lockout check	4.5.3.15.2		X
Hoisting cable check	4.5.3.15.3		X
Boom swing test	4.5.3.15.4		X
Boom elevation test	4.5.3.15.5		X
Hoisting hook position check	4.5.3.15.6		X
Rated capacity check	4.5.3.15.7		X
Radio interference suppression test	4.5.3.16		X

4.5.1.2 Fuel system, and hydraulic system check. To determine conformance to 3.4.2.1, 3.4.3 and 3.4.3.1, prior to initial fueling and filling, the fuel tank and hydraulic reservoir shall be checked for cleanliness.

4.5.1.3 Fuel system leakage check. To determine conformance to 3.4.2.1, prior to engine installation, the fuel system shall be checked for leakage.

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4..5.1.4 Hydraulic fluid leakage check To determine onformance to 3.4.3.2, the hydraulic system shall be operated at operating pressure and checked for leakage.

4.5.1.5 Exhaust system check. To determine conformance to 3.4.4, during loader operation the exhaust system shall be checked for functional requirements.

4.5.1.6 Adjustment mechanisms check. To determine conformance to 3.4.5, all adjustment mechanism settings shall be checked for proper adjustment and readjustment made when necessary.

4.5.1.7 Fastener checks. To determine conformance to 3.4.6, all fasteners, locks, and lights shall be operated during and after break-in run and checked for functional requirements.

4.5.1.8 Welding and burning check. To determine conformance to 3.4.7, all welding and burning operations shall be checked for removal of slag and if approved procedures have been followed.

4.5.1.9 Structural interferences check. To determine conformance to 3.4.10, during loader operation all specified items shall be checked for detrimental interferences.

4.5.1.10 Superstructure alinement check. To determine conformance to 3.4.13 the empty loader shall be indexed to the 10879601 alinement checkout fixture and superstructure shall be extended until hoisting beams are in their latching position. Operate the hoisting beam latch levers until the latches are properly engaged in the fixture pockets. The latches shall engage pockets readily without application of external lateral force to the hoisting beams. The manual latch shall release readily without evidence of binding.

4.5.1.11 Loader performance mobility tests. To determine conformance to 3.6 and 3.6.1, the loader shall be operated as specified, and examined for defects or malfunction.

4.5.1.11.1 Power package and power train test. To determine conformance to 3.6.2 and 3.6.3, the loader shall be operated throughout all ranges of operation and loader observed for functional requirements.

4.5.1.11.2 Speed tests. To determine conformance to 3.6.4, the loader shall be operated as specified and observed for speeds attained.

4.5.1.11.3 Creeping test. To determine conformance to 3.6.5, the loader shall be operated at a creeping speed and checked for distance of movement in specified time.

4.5.1.11.4 Acceleration test. To determine conformance to 3.6.6, each loader shall be given an acceleration test and checked for time required to attain specified speed.

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4.5.1.11.5 Climbing test. To determine conformance to 3.6.7, the loader shall be operated as specified, on specified grade, and checked for functional requirements.

4.5.1.11.6 Stopping test. To determine conformance to 3.6.8.1, the loader shall be checked for stopping requirements.

4.5.1.11.7 Holding test. To determine conformance to 3.6.8.2, loader standing on a 40 percent grade, with engine stopped for not less than one minute with parking brakes applied, the loader shall be held stationary.

4.5.1.11.8 Turning test. To determine conformance to 3.6.9, and 3.6.9.1, loader shall be operated in forward and reverse to the right and to the left in full 360 degree turns. The force required to start, turn, and maintain turn at all temperatures down to 32°F. shall be checked for conformance. The loader shall be operated in forward and reverse with brakes released and observed for turning to the right and left.

4.5.1.12 Hydraulic pressure at loader engine start-up test. To determine conformance to 3.6.18.1, the starter switch and engine shall be operated as specified and the hydraulic oil pressure gage checked for pressure requirements.

4.5.1.13 Loader electrical system check. To determine conformance to 3.6.18.2 through 3.6.18.10, the electrical system shall be operated and checked for functional requirements. The system shall be checked for completeness and mounting, and for any other defect that may cause failure of the electrical system.

4.5.1.14 Controls checks. To determine conformance to 3.6.19.1 through 3.6.19.5, the controls shall be operated as specified for specific controls and checked for functional requirements.

4.5.1.15 Hydraulic system test. To determine conformance to 3.6.20.1 and 3.6.20.2, the hydraulic pump shall be operated for not less than 15 minutes and checked for operating pressures. Relief and proof pressures shall be checked using an external pressure source and gages. Burst pressures shall not be applied.

4.5.1.16 Superstructure operation (normal loader operation) test. To determine conformance to 3.6.21, loader superstructure shall be operated as specified and checked for functional requirements.

4.5.1.16.1 Missile release circuits test. To determine conformance to 3.6.21.1, the hoisting beams shall be placed in their latched positions, with no missiles aboard, and a testing device similar to the one depicted in figure 2 schematic, shall be plugged into the receptacle on the end of the index boom. The wiring harness which has been detached from the index boom, and the wiring harness which supplies main power to the instrument box, shall be plugged into the testing device and the following check shall be made:

(a) Position the loader superstructure so that the index

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boom and transfer arm limit switches are closed (plungers extended) and place the master switch in the "ON" position.

(b) Place the Selector Switch in the "R" position. With the momentary switch on the test unit held in the closed position, the voltmeter shall read between 0.30 and 1.0 volt. A zero reading on the voltmeter indicates that the solenoid is not wired correctly.

(c) The test shall be repeated with the selector switch positioned on the remaining two positions.

4.5.1.17 Superstructure operating speeds test. To determine conformance to 3.6.21.2, the superstructure shall be operated as specified and observed functional requirements.

4.5.1.18 Superstructure settling test. To determine conformance to 3.6.21.3, with the vehicle superstructure loaded, the elevation and extension cylinders positioned as specified, and the engine off, superstructure shall be checked for settling.

4.5.1.19 Azimuth and roll controls limitation tests. To determine conformance to 3.6.21.4, the azimuth and roll controls shall be operated and their effectiveness observed.

4.5.1.20 Reduced pressure control test. To determine conformance to (a) through (d) of 3.6.21.5, the superstructure shall be operated as specified and observed for functional requirements.

4.5.1.21 Manual latching of missiles test. To determine conformance to 3.6.21.6, the manual latching and locking devices shall be operated as specified and checked for positive locking.

4.5.1.22 Manual release of missiles test. To determine conformance to 3.6.21.7, with the latch-release solenoid manually depressed, the missiles shall be checked to determine if fully released from the superstructure.

4.5.1.23 Forward missile support test. To determine conformance to 3.6.21.8, the outboard forward missile supports shall be operated as specified and observed for functional requirements.

4.5.1.24 Rear center missile support test. To determine conformance to 3.6.21.9, the rear center missile support and safety bar shall be stowed and checked for loader shipment and crane operation.

4.5.1.25 Superstructure electrical system checks. To determine conformance to 3.6.22 through 3.6.22.5, with the main cable connected as specified, the circuits shall be operated as specified and checked for functional requirement.

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4.5.1.26 Electrical release of missiles test. To determine conformance to 3.6.22.6, the missile latch-release shall be operated as specified and the electrical release checked for functional requirements.

4.5.1.26.1 Latch release solenoid test. To determine conformance to 3.6.22.6.1, with the hoisting beam assemblies in specified condition, the release solenoids shall be energized, and the latch locks and latching assemblies observed for functional requirements.

4.5.1.27 Index boom and transfer arm limit switches test. To determine conformance to 3.6.22.7, the limit switches shall be operatd as specified, and the solenoids and switches observed for functional requirements.

4.5.1.28 Loading and unloading missiles test. To determine conformance to 3.6.23 (a), (b) and (c), the loader shall load and unload the launcher and pallet on specified slopes and be checked for functional requirements.

4.5.1.29 Loading and unloading pallet test. To determine conformance to 3.6.24, the loader shall load and unload a pallet as specified and be checked for functional requirements.

4.5.1.30 Painting, marking, and data plates check. To determine conformance to 3.8.1, 3.8.2, and 3.8.3, the loader shall be checked to determine that identification, operational, and warning data plates and decals are correctly located and affixed and that painting and colors are as specified. Striping and indexing marks shall be in accordance with specified drawings.

4.5.2 Control test. Control tests for maintaining control of these requirements whose inspection would be of too long duration to be included under acceptance tests shall be conducted by the contractor as specified herein.

4.5.2.1 Frequency. The first loader off the production line, or one of the first ten loaders produced, and thereafter one per month, shall be subjected to the control tests as specified herein.

4.5.2.2 Test failure. If a vehicle fails to pass any control test specified herein, the Government inspector shall stop acceptance of subsequent lots until evidence has been provided by the contractor that corrective action has been taken.

4.5.2.3 Stowed material check. To determine conformance to 3.4.8, all manufacturers, depot, and troop on-vehicle-equipment (OVE) shall be stowed on a production vehicle. The OVE used shall be of the latest production available to the contractor, and the contractor shall be permitted to unpackage and shall repackage all OVE (requiring packaging) for domestic shipment only. The stowed OVE items shall be examined for installation and interference with the operation of the vehicle in any manner.

4.5.2.4 Missile transport test. To determine conformance to 3.4.11, the vehicle selected shall be subjected to a test of transporting specified number

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of missiles and checked for functional requirements.

4.5.2.5 Side slope test. To determine conformance to 3.6.10 the loader shall be operated on specified slopes and observed for functional requirements.

4.5.3 Comparison tests. The Government may select loaders any time during the contract production period and subject loaders to tests specified in 4.2.3 table VI to reveal deficiencies of a workmanship or materials nature that may reduce the effective operation of these items in the field and to compare existing quality with previous standards. These tests shall be conducted at Government laboratories or proving grounds designated by the contracting officer. Selection of loaders shall be on a spot-check basis. Comparison test loaders shall be fully loaded with all OVE stowed in their applicable spaces to conform to 3.4.8. Vehicles selected shall not include any loader previously tested for conformance to 4.5.2 (Control tests).

4.5.3.1 Test failure. Failure of any loader tested as specified herein to comply with any of the requirements specified in the contract, or any major deficiency of a workmanship or materials nature occuring, or as a result of the test, may be cause for refusal to continue acceptance of loaders by the Government until evidence has been provided by the contractor that corrective action has been taken to eliminate the condition.

4.5.3.2 Environmental tests. To determine conformance to 3.4.1, the loader, properly serviced and equipped, shall be subjected to applicable tests under conditions specified in Purchase Description MPD-7005.

4.5.3.3 Fuel system functional test. To determine conformance to 3.4.2, the loader shall be operated throughout all speed and gear ranges over cross country terrain and on relatively level, smooth, hard-surfaced roads, and loader observed for functional requirements. The system shall be checked for leaks before and after road test. After road test the fuel tank shall be examined for foreign matter.

4.5.3.4 Transportability test. To determine conformance to 3.4.9 and 3.7, the loader shall be subjected to such a test that will prove the transportability of the loader.

4.5.3.5 Lifting and towing test. To determine conformance to 3.4.12, the vehicle selected shall be subjected to a test of lifting and towing specified loads and observed for functional requirements.

4.5.3.6 Brake lever force test. To determine conformance to 3.6.9.1, the brake lever shall be applied under specified low temperature and checked for force required to operate brake lever.

4.5.3.7 Trench crossing test. To determine conformance to 3.6.11, the loader shall be operated across trenches of specified width and observed for functional requirements.

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4.5.3.8 Vertical obstacle test. To determine conformance to 3.6.12, the loader shall be operated over vertical obstacles of specified height and observed for functional requirements.

4.5.3.9 Blast hole test. To determine conformance to 3.6.13, the loader shall be operated through holes of specified diameter and depth, and observed for functional requirements.

4.5.3.10 Rough terrain test. To determine conformance to 3.6.14, the loader shall be operated over rough terrain as specified, and terrain as shown in figure 1.

4.5.3.11 Operation in snow test. To determine conformance to 3.6.15, the loader shall be operated in snow of specified depths and observed for functional requirements.

4.5.3.12 Fording test. To determine conformance to 3.6.16, the loader shall be operated in water as specified and observed for functional requirements.

4.5.3.13 Towing test. To determine conformance to 3.6.17, the loader shall be operated at specified speed, towing specified load, and observed for functional requirements.

4.5.3.14 Loading and unloading missiles test. To determine conformance to 3.6.23 (a), (b) and (c) the loader shall load and unload the launcher and pallet on specified slopes and be checked for functional requirements.

4.5.3.15 Crane operation test. To determine conformance to 3.6.25.1, the crane shall be operated as specified and checked for functional requirements.

4.5.3.15.1 Crane control tests. To determine conformance to 3.6.25.2, the controls shall be operated as specified and observed for functional requirements.

4.5.3.15.2 Roll lockout check. To determine conformance to 3.6.25.3, with roll-lockout lever positioned as specified, the roll lockout shall be checked for positive lockout.

4.5.3.15.3 Hoisting cable check. To determine conformance to 3.6.25.4, the extension control lever shall be operated as specified, the cable shall be observed for functional requirements.

4.5.3.15.4 Boom swing test. To determine conformance to 3.6.25.5, the operation of the azimuth control lever shall be observed for the functional requirements of the boom.

4.5.3.15.5 Boom elevation test. To determine conformance to 3.6.25.6, the operation of the elevation control lever shall be observed for the functional

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requirements of the boom.

4.5.3.15.6 Hoisting hook position check. To determine conformance to (a) and (b) of 3.6.25.7, the hoisting hook shall be positioned as specified and checked for distance from ground.

4.5.3.15.7 Rated capacity check. To determine conformance to 3.6.25.8, the loader used as a crane shall be operated as specified and observed for operational requirements. (Caution: Use crane sling to restrain load, and do not lift load more than one foot above ground.)

4.5.3.16 Radio interference suppression test. To determine conformance to 3.6.26, the vehicle shall be subjected to a radio interference suppression test.

4.5.4 Auxiliary equipment. The auxiliary equipment specified in 6.3, shall be used to check the compatability of the loader with this equipment.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, and packing. Preservation, packaging, and packing shall be in accordance with MIL-L-45779.

6. NOTES

6.1 Intended use. The loader-transporter is intended to transfer HAWK guided missiles from pallets to launcher and vice versa, and to otherwise handle missiles and pallets as may be necessary within a missile battery. It is also intended for limited use as a crane and for short distance towing.

6.2 Ordering data. Procurement documents shall specify the following:

(a) Title, number, and date of this document.
(b) Preproduction model if required (see 3.1).
(c) Stowing of On-Vehicle-Equipment if required (see 3.4.8).
(d) Simulated missile payload if required (see 4.5.1.1).

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6.3 Auxiliary	Drawing
Guided Missile, Air Defense: XM3E1 Standard Type	9071755
Launcher, Zero Length, Guided Missile:	9098880
Pallet, Loading and Storage, Guided Missile:	9196208
Truck, Cargo, 2-1/2 Ton, M36	8686878
Chassis, Trailer, 2 Ton, XM390	8736315
Ramp	9098330
Crane	9098300
Adapter, hoist	10916330

6.4 Definitions. For the purposes of this specification, the following definitions shall apply:

(a) Normal loader operation. Normal loader operation shall refer to operation associated with the loader's primary tactical function within a missile battery, namely that of transferring 1, 2, and 3 missiles at a time from storage pallets to launchers and vice versa, and transferring pallets from ground to XM390 trailers and the reverse. For such operation, indexing forks shall point toward front of loader, superstructure shall respond to hydraulic controls as specified in 3.6.21 and superstructure electrical circuits operate in accordance with 3.6.22. Normal operation shall involve indexing, positioning of superstructure on pallets or launchers for loading and unloading of missiles, latching of missiles to superstructure and release therefrom, transport of missiles latched to superstructure, and latching to, transporting and unloading empty pallets on XM390 trailers or the reverse.

(b) Normal loader transport position. Normal loader transport position shall refer to the position of the superstructure during missile transport by the loader (in normal loader operation) wherein the latched missile(s) are held firmly in loader saddles with full weight of each missile carried by the applicable front and rear loader missile supports.

(c) Final extension phase. Final extension phase shall refer to operation of superstructure in extension and retraction (in) normal loader operation) during period when reduced extension

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pressure feature is operative as specified in 3.6.21.5. In missile transfer operations, loader is indexed to launcher or pallet during the final phase of extension.

(d) Crane operation. Crane operation shall refer to operation of the loader when loader is rigged with items as specified (including ramp) and adjusted for crane operation. Loader shall be adapted as follows for crane operation:

1. Place forward and rear center missile supports in stowed positions (see 3.6.21.8 and 3.6.21.9).
2. Lockout carrier suspension system (see 3.6.19.2).
3. Unpin azimuth cylinder rod boom support, rotate boom through 3,200 mils and repin to azimuth cylinder.
4. Place roll lockout in "crane" position (see 3.6.25.3).
5. Extend superstructure until transfer arms contact link assembly and pin together using links provided.
6. Assemble crane attachments to ramp. Place ends of ramp in crane support bracket and pin.
7. Place crane support bracket on hoisting support boom trunnion shaft. Using quick release pins, pin front stays to boom support assembly and pin fairlead sheave assembly to center hoisting beam housing.
8. Using loader power, lift ramp with crane cable and pin rear cable stays to loader index forks.

6.5 The following shall be used with the classification of defects for leaks:

(a) Weep: Any evidence of fluid beyond the seal.
(b) Seep: Any evidence of fluid beyond the seal that does not result in formation of a droplet.
(c) Droplet: Any evidence of fluid beyond the seal that results in the formation of a droplet.
(d) Drip: Any evidence of fluid beyond the seal where droplet forms and falls.

6.6 Definitions of recurring major and minor defects. The Government Inspector will verify that a thorough inspection of **each vehicle is performed by the contractor not only for the listed characteristics, but also** for any other departures from good workmanship. The Government Inspector will assure that all deficiencies encountered during the inspection are enumerated on the Deficieny Sheet for the vehicle. The defects noted on Deficiency Sheet shall contain sufficient description to enable the Government Inspector and the contractor's representative to classify the deficiency in accordance with the classification of defects of the vehicle specification and definitions contained in MIL-STD-105. Corrective action shall be taken for recurring deficiencies.

Custodians:
Army - AT

Reviewer:
Army - MI

Preparing activity:
Army - AT

Project No. 1450 - A034

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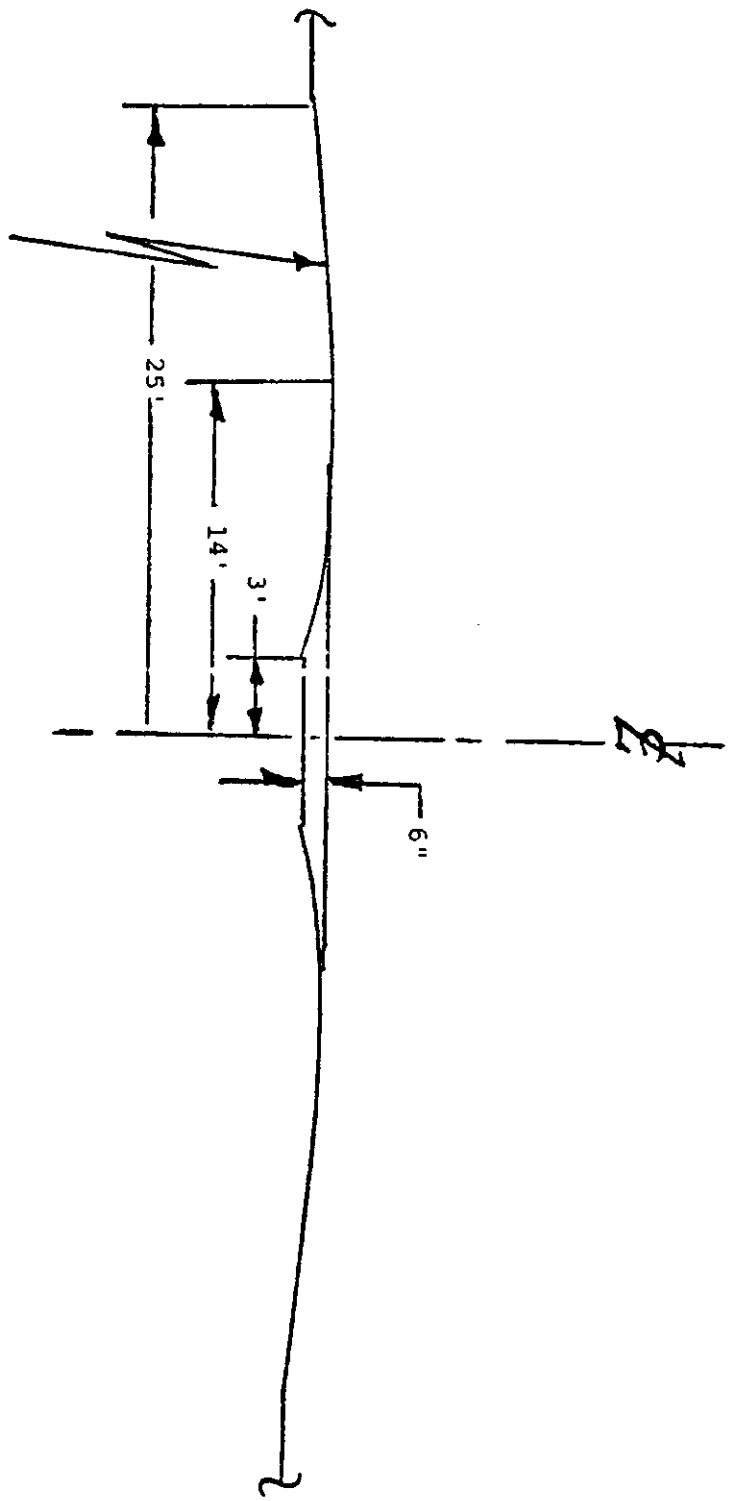


FIGURE 1 - PROFILE OF TEST TRACK
(ARTIFICIAL BUMP)

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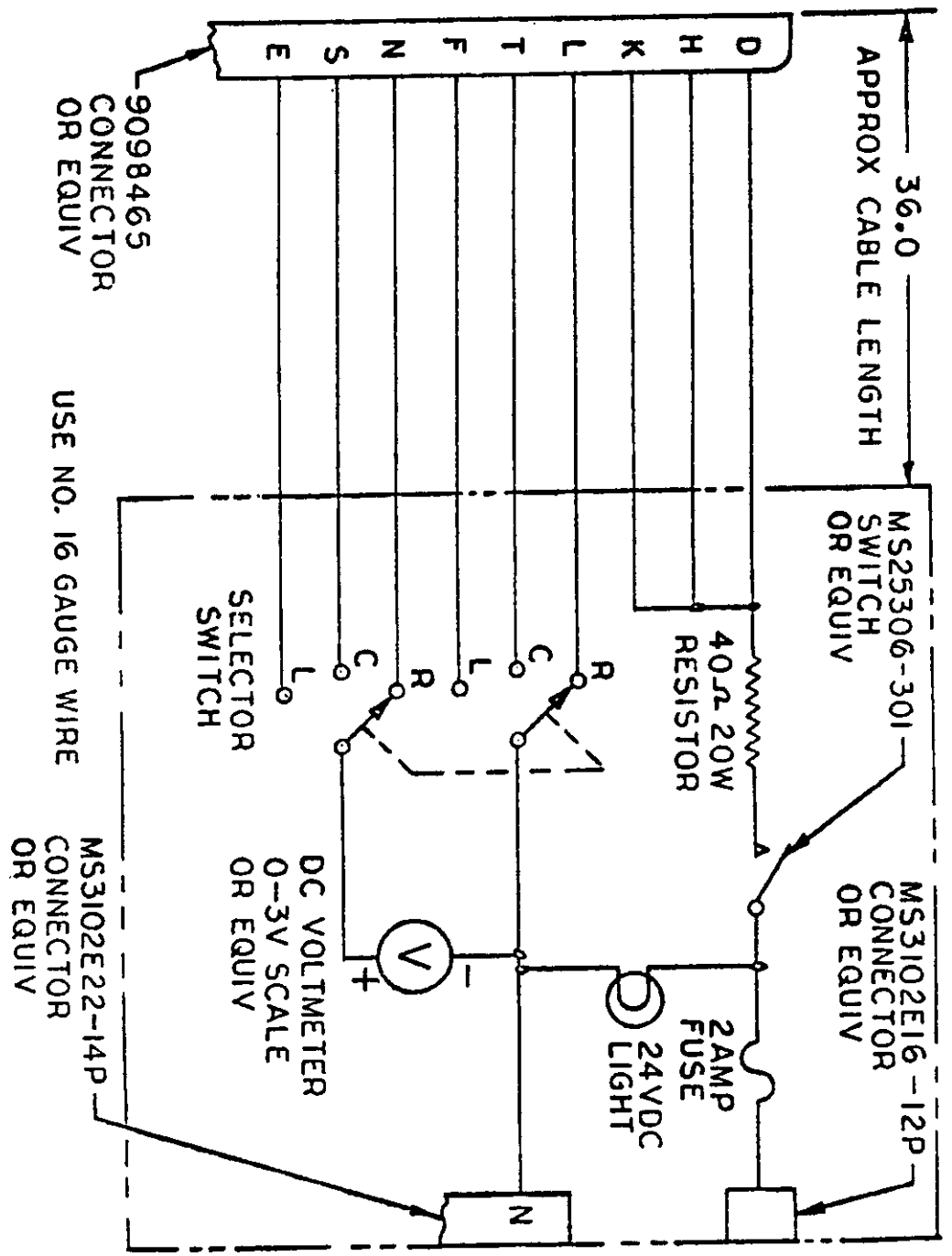


FIGURE 2

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US Army Tank-Automotive Command
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Warren, MI 48090

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

MIL-L-62174(AT)

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 54 WHICH MAY BE USED.

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